Stage 4 — Check Your Understanding

Q1. Why a 3D fibre needs the pressure balance

$$p(x) = G\cdot x + p'(x),\ x(s) = x0 + s\ t$$

$$p_{out}(s) = G\cdot x0 + s(G\cdot t) + p'(x(s))$$

The nonperiodic outside-pressure change per unit fibre length is G·t. With no interior body force, stationary fluid obeys grad(p_in) = 0, so its pressure is constant. The resulting pressure jump grows linearly along an aligned fibre, while periodic surface forces can support only a repeating traction jump. Adding b = G inside the fibre gives grad(p_in) = G and cancels the growing jump; for transverse fibres t·G = 0, so no such correction is needed in 2D. At epsilon = 0.2a the regularized solver still has axial interior velocity -0.117U, whose flux Eq. (7.10) removes.

Q2. Darcy’s law as a matrix

$$K^{-1} = (I - tt^T)/k_{perp} + tt^T/k_{parallel}$$

$$z = (t - e)/\sqrt{2},\ t\cdot e = 0,\ (z\cdot t)^2 = 1/2$$

$$z\cdot K^{-1} z = [1 - (z\cdot t)^2]/k_{perp} + (z\cdot t)^2/k_{parallel}$$

$$1/k_{tilted} = 1/(2\ k_{perp}) + 1/(2\ k_{parallel})$$

The test imposes superficial velocity U d and measures the required pressure-gradient component along d. Since G = -mu K^-1 U_sup, its measured directional permeability is 1/(d·K^-1 d), not d·K d. Resistances, rather than permeabilities, combine in this flow-controlled test. With the Table 6 alpha = 0.1 values, k_perp = 1.2652a² and k_parallel = 2.5531a², resistance averaging gives 1.6904a², versus 1.9092a² by arithmetic averaging (12.9% higher). Stokes reciprocity requires K = K^T, which the code does not enforce.

Q3. When Richardson extrapolation stops working

$$k(e) = k0 + c2\ e^2 + c3\ e^3 + O(e^4)$$

$$k(2e) = k0 + 4c2\ e^2 + 8c3\ e^3 + O(e^4)$$

$$4k(e) - k(2e) = 3k0 - 4c3\ e^3 + O(e^4)$$

$$k_R = [4k(e) - k(2e)]/3 = k0 - (4/3)c3\ e^3 + O(e^4)$$

The coarse width 2e must also be sufficiently small for this expansion to hold; narrow gaps in random layouts can violate that requirement. The maximum random-layout errors for pairs (0.2a,0.4a), (0.25a,0.5a), (0.3a,0.6a) are respectively 0.53%, 1.07%, 2.12% at alpha = 0.10; 0.44%, 0.88%, 1.85% at alpha = 0.15; and 1.01%, 2.10%, 4.42% at alpha = 0.20. Thus random networks are limited to alpha <= 0.15, and every aligned network is compared with its exact 2D answer to validate accuracy on that geometry.

Q4. The orientation rule against the 3D networks

$$1/k(c) = (1-c)/k_{perp} + c/k_{parallel}$$

$$\text{alpha} = 0.1:\ k_{perp} = 1.4977\ a^2,\ k_{parallel} = 3.7214\ a^2$$

$$c = 0:\ k_{through} = 1.4977\ a^2$$

$$c = 1/2:\ k_{plane} = 2(1.4977)(3.7214)/(1.4977+3.7214)\ a^2$$

$$k_{plane} = 2.1358\ a^2;\ \text{predicted ratio} = 2.1358/1.4977 = 1.4261$$

```
measured ratio = 1.33 ± 0.07; excess = (1.4261/1.33-1)*100 = 7.22%
```

Table 7 measured/rule ratios with bootstrap 95% intervals: layered through 1.061 [1.008,1.116] (alpha 0.10) and 1.114 [1.050,1.194] (0.15); layered in-plane 0.976 [0.944,1.010] and 1.061 [1.024,1.102]; cubic 0.949 [0.924,0.974] and 1.042 [1.014,1.071]. A ratio above 1 means underprediction, and below 1 overprediction. The layered in-plane alpha 0.10 interval includes 1, so its difference is not established.

For filter design, the rule correctly predicts greater in-plane than through-plane permeability, but its 1.43 anisotropy estimate exceeds the observed 1.33 by 7.2%. Across all six cases, measured/rule ratios range from 0.949 to 1.114, with five bootstrap intervals excluding 1. The rule underpredicts through-mat permeability, plausibly because crossing fibres create additional three-dimensional routes for air; this mechanism is not yet tested. The 12 aligned-network samples also show appreciable small-sample scatter.

Q5. One blob per fibre versus a ring of blobs

```
phi_3D = exp[-(r_perp²+z²)/e²] / (pi^(3/2) e³)

Integral[-infinity,infinity] phi_3D dz = exp[-r_perp²/e²]/(pi e²)
```

Equation (7.6) therefore reduces a uniform line of 3D Gaussian blobs exactly to the 2D Gaussian blob in each cross-section. Proposition 3 gives effective cylinder radius a_eff = e_c exp(-gamma/2). Requiring a_eff = a yields e_c = a exp(gamma/2) = 1.3346a, which matches the dilute far-field cylinder response.

The cheap centreline model is useful for preliminary permeability or flow estimates in widely spaced arrays where its error has been checked. It is not reliable for quantitative particle interception in Stages 5–6 because interception depends on near-surface flow. Table 7 gives mean centreline errors of +1.8% to +4.4%, with individual random networks up to 12% off at alpha = 0.15; use resolved rings for capture calculations.